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Evaluating efficacy of treatments for erythromelalgia in the setting of biopsy proven small fibre reduction

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Dear Editor,

Erythromelalgia is characterized by redness, pain and elevated temperature, and probably reflects dysfunction of small neurite fibres of the skin. Treatment is often challenging and although erythromelalgia is associated with inflammatory conditions, it is unclear whether the best therapy involves immunosuppression or neuromodulatory therapy, especially in cases where the disease is advanced and neurite fibre decline has occurred.^{1–3} The aim of this study was to assess the response to treatments in patients with erythromelalgia with biopsy-proven reduced nerve density in the skin.

Using the Partners Healthcare Research Patient Data Registry, we identified 352 patients with epidermal neurite density (END) less than the fifth percentile for their age and sex who were seen at the Brigham and Women's Hospital and the Massachusetts General Hospital from January 2000 to December 2019. Individual chart review resulted in the identification of 26 clearly documented patients with erythromelalgia, whose prescription histories were individually assessed. Treatment was considered successful if improvement of erythromelalgia presented was reported by the attending physician at a follow-up visit.

The baseline characteristics of patients included are presented in Table 1. Although all patients with erythromelalgia had substantially decreased END (median percentile for age and sex 1.05; interquartile range 1.2), concomitant pain was a symptom very frequently seen (22 of 26 patients; 84.6%). Drugs aimed at neuromodulation (gabapentin, pregabalin, tricyclic antidepressants, topiramate, carbamazepine) generally produced the lowest responses (< 50%) although, of these, the selective serotonin reuptake inhibitors resulted in better control (5 of 9; 55.6%) than did other medications in this category (Table 2). By contrast, higher responses were seen for patients prescribed treatments with immunomodulatory effects, including intravenous immunoglobulin (7 of 11; 63.7%), systemic corticosteroids (6 of 9; 66.7%), and rituximab (3 of 4; 75%). Of note, pharmacological sodium channel blockade with mexiletine was used in three of four cases (75%) and in particular, was beneficial for patients with an identified SCN9A mutation (2 of 2; 100%).

Table 1 Baseline characteristics of patients included ($n = 26$).

Characteristic	Value (%)
Age, years; median (IQR)	46.5 (21.25)
Sex, n (%)	
Male	6 (23)
Female	20 (77)
Skin biopsy findings	
END, neurites/mm ² ; mean $\pm$ SD	102.3 $\pm$ 66.4
Percentile; median (IQR) ^a	1.05 (1.2)
Alternative aetiologies, n (%)	
Autoimmune	12 (46)
Idiopathic	5 (19)
Genetic/hereditary	3 (11.5)
Sjögren syndrome	3 (11.5)
Fibromyalgia	3 (11.5)
Lupus erythematosus	1 (4)
Sarcoidosis	1 (4)
Lyme disease	1 (4)

END, epidermal nerve density; IQR, interquartile range. ^aAge and sex.

Table 2 Treatment response rates.

Drug	Users, n	Improved, n	Response rate, %
Botulinum toxin	2	2	100
Rituximab	4	3	75
Mexiletine	4	3	75
Systemic steroids	9	6	66.7
IVIg	11	7	63.7
SSRI	9	5	55.6
TCA	10	5	50
Gabapentin	9	4	44.4
Pregabalin	8	3	37.5
Topiramate	5	1	20
Carbamazepine	1	0	0

IVIg, intravenous immunoglobulin; SSRI, selective serotonin reuptake inhibitor; TCA, tricyclic antidepressant.

Although our pilot study is limited in that it represents the findings from a single academic centre, we have a high-quality dataset with biopsy-proven nerve density changes. Our findings show that although systemic medications in general did not produce satisfactory improvement (< 70%), drugs with anti-inflammatory properties were more effective than those targeting neuromodulation; however, there may be a role for both.

This finding may be clinically important as it also suggests that imbalanced inflammation may play a more relevant role in the clinical symptoms of erythromelalgia than the dysfunction of neurite fibres in the skin, in which case drugs with immunosuppressant properties should be selected even when a solid decline in epidermal neurite fibres is found. In particular, sodium channel blockade with mexiletine may be the most helpful approach especially for the subset of patients identified with an SCN9A-related inherited disorder.

Conflict of interest

The authors declare that they have no conflicts of interest.

Funding

None.

Ethics approval

This study was approved by the Mass General Brigham Institutional Review Board 2019P003357/MGH. The patient has provided informed consent to publication of their case details and images.

Data sharing

Data are available on request from the authors.

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COVID-19 and favipiravir induce cutaneous adverse reactions: a difficulty in clinical dermatology practice

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Linked article: Punyaratabandhu P, Vanitchpongphan S. *Clin Exp Dermatol* 2022; **47**: 573–7.

Dear Editor,

We read with interest the recent article by Punyaratabandhu and Vanitchpongphan in *Clinical and Experimental Dermatology* describing cutaneous adverse reactions (CARs) in patients with COVID-19 treated with favipiravir.¹ The authors stated that in such patients, medication allergy should be considered and excluded as a cause of the rash.¹ We agree with the authors that favipiravir can have CARs. A recent examination of the efficacy of favipiravir, lopinavir/ritonavir and arbidol found identical efficacy between the drugs, but a much higher rate of



Figure 1 Maculopapular rash occurring 10 days after starting a 2-week course of favipiravir.

CAR with favipiravir.² However, such lesions were not generally subjected to specific pathological testing, and the likelihood of unfavourable consequences is based solely on speculation. A patient with COVID-19 may have a cutaneous issue despite receiving no treatment,^{3,4} and therefore other possible causes of rash, such as other medications, pre-existing personal sickness and concomitant medical problems must be excluded. There is a clinical challenge is distinguishing between skin rash caused by COVID-19 and that caused by favipiravir. Although a previous report¹ described careful exclusion of other causes of other concurrent factors, there was no confirmation by skin test.

We report a similar case, that of a 28-year-old woman who developed a maculopapular rash on her limbs (Fig. 1), which began on Day 10 of a 2-week course of favipiravir 30 mg/kg/day for COVID-19 infection. The patient had previously been well and was not taking any other medication. A skin prick test for allergic reaction to favipiravir yielded a positive result.

Favipiravir was continued for the remaining 4 days (Days 11–14) of the 2-week course and the patient was also started on oral chlorpheniramine 12 mg/day. There was complete clearance following 7 days of